

be very difficult to add them later on.

On the other hand, things such as MMC/SD cards, character LCDs, serial interfaces, LEDs and switches are easy to add and are not so much of a big deal to wire up. But it is better to get a board that has them so that you do not waste valuable time investigating failures caused due to soldering skills.

FPGAs are quite a complex device and, to build a project on a development board, it is important that you know how the interfaces and other hardware is wired on the board. Most manufacturers have proper legends on the board to indicate the interfaces but wiring sometimes can be difficult to trace. So check that the manufacturer provides complete schematic diagram, a reference manual, getting started guide, reference designs and sample codes.

There are many vendors selling development boards around FPGA chips from the manufacturers mentioned in the table. Below are some boards of different FPGA families:

Development boards for Xilinx chips

Spartan-6

1. Digilent Nexys 3

- Xilinx Spartan6 XC6SLX16-CS324 FPGA
- 16MB Micron Cellular RAM
- 16MB Micron Parallel PCM
- 16MB Micron Quad-mode SPI PCM
- 10/100 SMSC LAN8710 PHY
- Digilent Adept USB port for power, programming and data transfers
- USB-UART
- Type-A USB host for mouse, keyboard or memory stick
- 8-bit VGA
- 100MHz fixed-frequency oscillator



Fig. 2: Digilent Nexys 3 development board

- Eight slide switches, five push buttons, 4-digit seven-segment displays and eight LEDs
- Four double-wide Pmod connectors and one VHDC connector
- Rugged plastic case and USB cable included

2. The Mojo

- Spartan 6 XC6SLX9 FPGA
- 84 digital I/O pins
- Eight analogue inputs
- Eight general-purpose LEDs
- One reset button
- One LED to show when the FPGA is correctly configured
- On-board voltage regulation that can handle 4.8V-12V

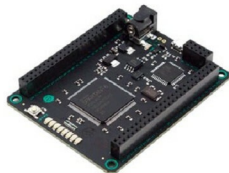


Fig. 3: Mojo development board

- A microcontroller (ATmega16U4) used for configuring the FPGA, USB communications and reading the analogue pins
- On-board Flash memory to store the FPGA configuration file

3. ZTEK USB FPGA module

- High-Speed (480Mbps) USB interface via mini-USB connector (B-type)
- Cypress CY7C68013A EZ-USB FX2 microcontroller
- Xilinx Spartan 6 LX9 to LX25 FPGA (XC6SLX9 to XC6SLX25)
- 90 general-purpose I/O's (GPIOs)
- 15 special I/O's (SIO)
- 64MB DDR SDRAM
- MicroSD socket for extensible Flash memory (standard and high-capacity (SDHC) cards, SPI mode support required)
- Bitstream loading from Flash to FPGA supported by the firmware
- Accessible from EZ-USB FX2, from FPGA and from host software using the SDK



Fig. 4: ZTEK USB FPGA module

- 128 kbit EEPROM memory (can be used to store the EZ-USB firmware)
- FPGA configuration/programming via USB using the EZ-USB FX2 SDK. (No JTAG adaptors or other additional utilities required)
- FPGA configuration/programming from Flash memory

ZTEK USB FPGA module has no on-board LEDs or buttons, but contains 64MB of SDRAM and a huge 90 GPIOs in a convenient plug-in module form factor, which is nice for dropping into a custom carrier board. Unfortunately, it does not have its own built-in power supply, so you need to place it onto the power supply or experimenter board, and then drop that onto your own board.

4. Avnet Spartan-6 Microboard

- Spartan-6 XC6SLX9-2CSG324C FPGA
- 64MB LPDDR SDRAM
- 128Mb multi-I/O SPI Flash
- 10/100 Ethernet PHY
- USB-to-UART port
- On-board USB JTAG circuitry
- Two Digilent Pmod compatible headers (2x6)
- Single-chip, 3-rail power with Power Good indicator
- Programmable clock chip
- Over-voltage and ESD protection on USB
- Four LEDs
- 4-bit DIP switch
- Reset and PROG push-buttons

Avnet Spartan-6 Microboard appears to be similar to the Arrow BeMicro. It is designed for embedded processor development and comes with a



Fig. 5: Avnet Spartan-6 Microboard